Final Project – Birds Biodiversity Temporal Trends

Overview

This capstone project asks you to investigate the long-term monitoring data collected in the Birds Biodiversity programme. Your objective is to quantify how biodiversity indicators have evolved, provide sound statistical uncertainty assessments, and highlight species-specific stories that emerge from the dataset.

The brief outlines **what** must be addressed without prescribing **how** you should proceed. Choose statistical methods, visualisations, and modelling strategies that you consider appropriate, and justify every major decision.

You will work with the STOC monitoring extract provided at `projects/birds-biodiversity/data/raw/Observations 2012-2025.xlsx`. Build your own helper code or reuse any open-source statistical libraries you deem suitable.

Core Deliverables

1. **Technical report** (PDF or Markdown) summarising your approach, indicators studied, and key findings.

2. **Reproducible analysis code** (notebook(s) and/or scripts).

3. **Figures and tables** supporting your conclusions.

Key Questions to Address

1. Dataset Familiarisation and Descriptive Analysis

- Describe the dataset structure (dimensions, key columns, time span, identifiers such as transects/observers/species).

- Produce descriptive statistics showing how the main variables are distributed. Pay particular attention to observer effort because it influences the interpretation of all subsequent results.

- Discuss data quality considerations (missing values, outliers, anomalous periods, etc.).

2. Multi-Year Indicator Trends

- Choose at least **three** biodiversity or sampling indicators you consider informative. Explain your selection.

- Compute annual estimates for 2014–2025 and supply confidence intervals (method and level are up to you, but justify the choice).

- Quantify and interpret temporal trends using appropriate models or descriptive tools.

3. Species-Level Evolution

- Select a subset of species and examine how their counts/presence evolve over time.
- Provide uncertainty assessments and discuss ecological or operational explanations for the patterns you observe.

4. Synthesis and Recommendations

- Summarise insights about the monitoring programme, highlighting converging or diverging evidence across indicators.
- Provide recommendations for future data collection or management where possible. Lack of clear actions will not be penalised, but sloppy analysis will.
- Reflect on limitations (data quality, modelling assumptions, sensitivity analyses).

Guidance

- The aim is to demonstrate statistical creativity and rigour rather than to discover groundbreaking ecological phenomena.
- Novel indicators or visualisations are welcome if they are well motivated and correctly implemented.
- Incorrect conclusions driven by flawed assumptions or buggy code will be penalised heavily. Validate every step.
- Use standard statistical libraries as needed, but you are accountable for every line of code.
- Keep visualisations polished. If warnings remain, explain why they cannot or should not be suppressed.
- Comment your code thoroughly so reviewers can understand your reasoning without guesswork.

Suggested Workflow

1. Explore and clean the data.
2. Define indicators and species of interest with justification.
3. Implement analyses (structure notebooks/scripts clearly).
4. Validate results (diagnostics, sensitivity checks, reproducibility runs).
5. Assemble the final report, figures, tables, and README.

Evaluation Criteria

- **Statistical soundness**
- **Insight and interpretation**
- **Clarity and reproducibility**

- **Creativity and initiative**

Submission

- Package all deliverables (report, code, figures, tables, README) into a single archive named `FinalProject_GroupName_Lastname1_Lastname2[_Lastname3].zip`.

- Email the archive to `stephane.rivaud@universite-paris-saclay.fr` **before 6 November, 23:59 (Paris time)**. Late submissions are not accepted.

- The README must describe the archive contents and provide reproduction instructions (environment, entry point, runtime notes).

- You may document the study in a PDF report or in notebook form (ensure notebooks are self-contained).

- Each group (2 or 3 students) must list all members and student IDs in both the README and the report/notebook.

- Oral defences will take place between 7 and 14 November (10-minute presentation + 5-minute Q&A;).

Good luck—use this project to showcase the statistical toolkit you have acquired!